EXPLORING THE LETHALITY & CHEMISTRY OF

Cyanide

[redacted]

IUPAC Name:
Hydrogen cyanide

Molecular Formula:
HCN (with N≡C-)

Molar Mass:
27.03 g/mol

CYANIDE INFORMATION

$$H-C\equiv N$$

FUNCTIONAL GROUPS & KEY FEATURES:
- triple-bonded carbon-nitrogen (C≡N) group
- extremely small size for high permeability
- strong affinity for metal ions (esp. Fe)

Key component of many **TOXIC SUBSTANCES!** Known for rapid, deadly action.

1. Toxicity & Action
- Potent cellular asphyxiant
- Rapidly binds to cytochrome c oxidase in mitochondria
- Inhibits aerobic respiration, leading to 'cell suffocation'
- Common symptom: cherry-red blood

2. Essential Chemical Properties
- Colourless, extremely volatile gas (HCN) or crystal salt (NaCN)
- Odour often described as 'bitter almonds' (but many people cannot smell it)
- Extremely toxic upon inhalation, ingestion, or skin contact

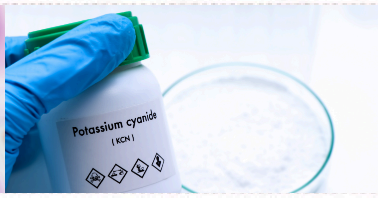
3. What Makes it So Dangerous?
- Fast-acting (death in minutes for high doses)
- Ubiquitous industrial use makes it a constant risk
- Historical use in chemical warfare (e.g., Zykion B) and suicide pills

4. Creative & Industrial Uses
- Primarily used in gold and silver mining (leaching process)
- Production of acrylic fibers and plastics
- Synthesis of chemical intermediates (e.g., nitriles)

Note: Uses are industrial, not creative in the traditional sense.

5. Sources & Exposure
- Natural sources: cassava, some fruit pits (as amygdalin), volcanic gases
- Industrial sources: manufacturing (e.g., electroplating), mining
- Common source of low-level exposure: tobacco smoke

6. Safety & Decontamination
- Use of specialized PPE (respirators, suits)
- Rapid detection systems
- Post-exposure antidotes exist (e.g., hydroxocobalamin, thiosulfate)



APA CITATIONS

1. National Center for Biotechnology Information (NCBI). (2024). *PubChem Compound Summary for CID 768, Cyanide*. Bethesda (MD): National Library of Medicine (US).
2. Centers for Disease Control and Prevention (CDC). (2023). "Specific Chemical Information: Cyanide". *Emergency Preparedness and Response*. (Specifically the section on exposure and toxicity).

Safety & Risk Note

SAFETY NOTE: Cyanide compounds are extremely toxic and dangerous. Handle with masimum care in professional settings.
RISK: Accidental release can lead to mass casualty events.